APART FROM 9, EVERY DENOMINATOR OF $1/(2-3/(3-4/(\cdots(n-1)-n/(-3))))$ IS 1 OR A PRIME

MATH0

Abstract. For $n \geq 3$ let $a(n)$ be the denominator, in lowest terms, of the finite continued fraction $1/(2-3/(3-4/(\cdots(n-1)-n/(-3))))$; this is the OEIS sequence A362086. We prove the conjecture of M. Bouras (2023) that every term other than 9 is 1 or a prime, and the argument shows that $a(5) = 9$ is the only composite term. The proof derives the closed form $a(n) = N/\gcd\big(N, 3(n-1)(n-3)!\big)$ with $N = n^2+n-3$ from the convergents of the continued fraction, studies the least prime factor of $a(n)$ for $n \geq 36$, and reads off the cases $n \leq 35$ from the listed terms. To our knowledge this is a new result: the entry records the conjecture without a proof, the analogous conjectures for the endpoints -1, -2, -4 and -5 were settled earlier, and our argument adapts the one used for the endpoint -5. The proof is verified in Lean 4 against a statement that math0 formalized from the OEIS entry and checked against its 57 listed terms; the statement is not part of the formal-conjectures repository.

1. Introduction

In 2022–2023 M. Bouras contributed to the OEIS five sequences of denominators of the finite continued fractions $1/(2-3/(3-4/(\cdots(n-1)-n/(-c))))$, $c = 1, \ldots, 5$: A356247, A363102, A362086, A363347 and A363482 [14, 16, 15, 17, 18]. By the formulas in these entries the unreduced denominator is $n^2+(c-2)n-c$, and each entry conjectures that the reduced denominators are 1 or prime, with at most one exceptional value; see also [2, 3]. For $c = 3$ the unreduced denominator is $N = n^2+n-3$, and the entry A362086, created by Bouras on May 28, 2023, states: "Conjecture: Except for 9, every term of this sequence is either a prime or 1." We prove this conjecture (Theorem 2); we do not address the entry's further conjecture, due to B. McEachen, on record values.

Prior work. For $c = 1$, the conjectures of A356247 are proved in a note of B. Cloitre [4], according to a 2025 comment in that entry. For $c = 2$, the conjecture that A363102 contains only 1's and primes is marked as solved in the formal-conjectures repository [5], with a formal proof found by the methods of Tsoukalas et al. [11]; T. Adamczewski gives a proof sketch in [1, Appendix A], and the AlphaProof Nexus outputs [6] contain a Lean file for it. For $c = 4$, Conjectures 1 and 2 of A363347 were proved by an AI agent (AlphaProof Nexus) [6]; see also [1, Appendix A]. Conjecture 2 of A363347 is also marked as solved in formal-conjectures [5], with a formal proof found by the methods of [11]. For $c = 5$, the conjecture of A363482 (exceptional value 49) was proved, with a Lean proof, in the math0 project [8].

For $c = 3$ the math0 referees found no proof. The current revision of A362086 (#35, May 9, 2024) records the conjecture but no proof, and Bouras's preprint [3] lists A362086 in the family $n^2+(k-2)n-k$ of its Table 1 and states the primality property only as a conjecture. The sequence has no file in formal-conjectures [5] (at the pinned commit `c77fdd4`), and the referees found it neither among the AlphaProof Nexus outputs [6] nor in Epoch AI's LeanOpenProblems repositories. Searches of arXiv, MathOverflow and Mathematics Stack Exchange found no proof or counterexample. We therefore classify Theorem 2 as a *new result*, with two caveats. First,

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the method is not new: the proof is an adaptation of the math0 proof for $c = 5$ [8], which in turn follows the template of the $c = 2$ proof in [1, Appendix A] (a closed form for the unreduced fraction, then bounds on the powers of small primes dividing factorials). Second, the referees could not read Cloitre's note [4], so we cannot exclude that it treats a general endpoint $-c$; the entry A356247 describes it only as a proof of that entry's conjectures, and A362086 carries no reference to it.

2. Statement

For $q \in \mathbb{Q}$ we write $\operatorname{den} q \geq 1$ for its denominator in lowest terms.

Definition 1. For an integer $k \geq 2$ and $t \in \mathbb{Q} \setminus \{0\}$ put $s_k(t) = k - (k+1)/t$. For $n \geq 3$ let $t_n = -3$ and $t_k = s_k(t_{k+1})$ for $k = n-1, \ldots, 2$, and put

$$\operatorname{cf}(n) = 1/t_2 = 1/\big(2 - 3/(3 - 4/(\cdots(n-1) - n/(-3)))\big), \qquad a(n) = \operatorname{den}\operatorname{cf}(n).$$

That no t_k vanishes is shown in Proposition 6 (for $n = 3$, $t_2 = s_2(-3) = 3$). For example $\operatorname{cf}(3) = 1/3$, $\operatorname{cf}(4) = 13/17$, $\operatorname{cf}(5) = 13/9$ and $\operatorname{cf}(6) = 41/13$. The OEIS entry [15] (offset 3) is named "Denominator of the continued fraction 1/(2-3/(3-4/(4-5/(...(n-1)-n/(-3)))))", gives the first three examples, and contains the comment, verbatim,

> `Conjecture: Except for 9, every term of this sequence is either a prime or 1.`

which is unsigned and hence due to the author of the entry, Mohammed Bouras, May 28, 2023.

Theorem 2. *For every $n \geq 3$ with $a(n) \neq 9$, $a(n) = 1$ or $a(n)$ is prime.*

Corollary 3. *For $n \geq 3$, $a(n)$ is composite if and only if $n = 5$; in particular $a(n) = 9$ only for $n = 5$.*

So the two readings of "Except for 9" (an exception for the value 9, or for the term $a(5)$) are equivalent.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A362086.lean`, reviewed for fidelity and frozen (SHA-256 prefix `e9131261`) before any proof attempt. Its definitions and statement are, verbatim except that docstrings are omitted:

```
namespace Math0.Statements.A362086

def step (k : ℕ) (t : ℚ) : ℚ := (k : ℚ) - ((k : ℚ) + 1) / t

def cf (n : ℕ) : ℚ := 1 / (List.range' 2 (n - 2)).foldr step (-3)

def a (n : ℕ) : ℕ := (cf n).den

def statement : Prop :=
  ∀ n : ℕ, 3 ≤ n → a n ≠ 9 → a n = 1 ∨ (a n).Prime

end Math0.Statements.A362086
```

`step k t` is $s_k(t)$. For $n \geq 3$, `List.range' 2 (n - 2)` is the list $[2, 3, \ldots, n-1]$, and the right fold of `step` over it, starting at -3, is `step 2 (step 3 (...(step (n-1) (-3))))`, i.e. t_2. So `cf n` is $\operatorname{cf}(n)$, and `a n` is its reduced denominator `Rat.den`, which is positive. This is the definition in the name of the entry, not the gcd formula listed there. The statement quantifies over $n \geq 3$, the offset of the entry, and reads "Except for 9" as an exception for the value 9; values of `a n` for $n < 3$ are never used. Lean defines $q/0 = 0$, which would change the value if some t_k vanished; this never happens: $t_n = -3$, and for $n \geq 4$ Proposition 6 gives

$t_k \ge k-2 > 0$ for $3 \le k \le n-1$ and $t_2 > 0$ (for $n = 3$, $t_2 = 3$). The formal proof evaluates the fold under the same bounds (Lemma 5). The same file checks the definitions against the entry by kernel evaluation (`decide +kernel`): the theorem `a_data` states that $(a(3), \dots, a(59))$ is the list $[3, 17, 9, 13, 53, \dots, 367, 263, 131]$ of all 57 terms in the data section of the entry, and an `example` confirms the entry's examples $\mathrm{cf}(3) = 1/3$, $\mathrm{cf}(4) = 13/17$, $\mathrm{cf}(5) = 13/9$.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A362086.lean` (Section 4). Throughout, $x = n - 3 \ge 0$,

$$N = x^2 + 7x + 9 = n^2 + n - 3, \qquad M = 3(x+2)\,x! = 3(n-1)(n-3)!,$$

and $!\,m = \sum_{i=0}^{m-1} i!$ is the left factorial (A003422 [12]; the notation is Kurepa's [7]). Thus $!\,0 = 0$, $!\,1 = 1$, $!\,2 = 2$ and $!(m+1) = !\,m + m!$; as $m!$ is even for $m \ge 2$, $!\,m$ is even for every $m \ge 2$.

3.1. **Convergents and the closed form.** For $m \ge 0$ define the linear polynomials

$$P_m(t) = 2\big((m+2)t - (m+1)(m+3)\big), \qquad Q_m(t) = (m+2)\,!(m+1)\,t - (m+1)(m+3)\,!\,m,$$

and, where defined, $G_m(t) = s_2(s_3(\cdots s_{m+2}(t)\cdots))$. Thus $G_0 = s_2$ and $G_{m+1}(t) = G_m(s_{m+3}(t))$.

Lemma 4. *For $m \ge 0$ and $t \ne 0$, $t\,P_m(s_{m+3}(t)) = P_{m+1}(t)$ and $t\,Q_m(s_{m+3}(t)) = Q_{m+1}(t)$.*

Proof. Put $s = s_{m+3}(t)$, so $ts = (m+3)t - (m+4)$. Then

$$\begin{aligned} tP_m(s) &= 2\big((m+2)((m+3)t - (m+4)) - (m+1)(m+3)t\big) \\ &= 2\big((m+3)t - (m+2)(m+4)\big) = P_{m+1}(t), \\ tQ_m(s) &= (m+3)\big[(m+2)\,!(m+1) - (m+1)\,!\,m\big]t - (m+2)(m+4)\,!(m+1) = Q_{m+1}(t), \end{aligned}$$

because $(m+2)\,!(m+1) - (m+1)\,!\,m = !\,m + (m+2)\,m! = !(m+1) + (m+1)! = !(m+2)$. □

Lemma 5. *Let $m \ge 0$ and $t \in \mathbb{Q}$ with $t \ge m+1$. Then $G_m(t)$ is defined, the argument of each s_k $(2 \le k \le m+2)$ occurring in it is at least $k-1$, $Q_m(t) > 0$, and $G_m(t) = P_m(t)/Q_m(t)$.*

Proof. Q_m has positive leading coefficient and $Q_m(m+1) = (m+1)\big[(m+2)\,m! - !\,m\big] > 0$, since $!\,m \le m \cdot (m-1)! = m!$ for $m \ge 1$ and $!\,0 = 0$; hence $Q_m(t) > 0$. The rest is by induction on m. For $m = 0$, $t \ge 1$ and $G_0(t) = 2 - 3/t = (2t-3)/t = P_0(t)/Q_0(t)$, as $P_0(t) = 2(2t-3)$ and $Q_0(t) = 2t$. For the step let $t \ge m+2$. Then $(m+4)/t \le (m+4)/(m+2) \le 2$, so $s = s_{m+3}(t) \ge m+1$. By the induction hypothesis $G_{m+1}(t) = G_m(s)$ is defined, with the stated bounds on the arguments, and by Lemma 4

$$G_{m+1}(t) = \frac{P_m(s)}{Q_m(s)} = \frac{tP_m(s)}{tQ_m(s)} = \frac{P_{m+1}(t)}{Q_{m+1}(t)}.$$ □

In Lean, $P_m(t)/Q_m(t)$ is `F m t`, and Lemmas 4 and 5 are `F_step` and `fold_eq_F`; the positivity of Q_m is not needed there, as with $q/0 = 0$ we have $(ta)/(tb) = a/b$ for all a, b when $t \ne 0$.

Proposition 6. *Let $x \ge 1$, i.e. $n \ge 4$. Then $t_k \ge k-2 > 0$ for $3 \le k \le n-1$, $t_2 = 2N/(N \cdot !\,x + M) > 0$, and*

$$\mathrm{cf}(x+3) = \frac{N \cdot !\,x + 3(x+2)\,x!}{2N}.$$

Proof. Let $n = x+3$ and $u = t_{n-1} = s_{x+2}(-3) = (x+2) + (x+3)/3 \geq x$. Lemma 5 with $m = x-1$ gives the bounds on $t_3, \dots, t_{n-1}$ and $t_2 = G_{x-1}(u) = P_{x-1}(u)/Q_{x-1}(u)$. Lemma 4 with $m = x-1$ and $t = -3$ gives $-3P_{x-1}(u) = P_x(-3)$ and $-3Q_{x-1}(u) = Q_x(-3)$, and

$$P_x(-3) = 2\big(-3(x+2) - (x+1)(x+3)\big) = -2N,$$
$$Q_x(-3) = -3(x+2)\big(!\,x + x!\big) - (x+1)(x+3)\,!\,x = -\big(N \cdot !\,x + M\big).$$

So $t_2 = 2N/(N \cdot !\,x + M) > 0$, and $\mathrm{cf}(x+3) = 1/t_2$ is as stated. □

For $x = 1, 2, 3$ the formula gives $26/34$, $78/54$ and $246/78$, in accordance with the examples; it also holds for $x = 0$, where it gives $6/18 = 1/3$, but the formal proof does not need this case. In Lean, Proposition 6 is `cf_closed`, stated with $x = y + 1$.

3.2. **The reduced denominator.**

Proposition 7. *N is odd, and for $x \geq 2$, $a(x+3) = N/\gcd(N, M)$.*

Proof. $N = x(x+1)+6x+9$ with $x(x+1)$ even. For $x \geq 2$ both $!\,x$ and $x!$ are even, so $N \cdot !\,x + M = 2X$ with $X \in \mathbb{N}$, and $\mathrm{cf}(x+3) = X/N$ by Proposition 6. Hence $a(x+3) = N/\gcd(N, X)$, and since N is odd, $\gcd(N, X) = \gcd(N, 2X) = \gcd(N, M + !\,x \cdot N) = \gcd(N, M)$. □

In Lean these are `N_odd` and `a_eq`; the latter is stated for $x \geq 3$, which is all that is used. The entry also gives the formula $a(n) = N/\gcd\big(N, 3\,A051403(n-3) + n\,A051403(n-4)\big)$, where $A051403(m) = (m+2)\,!(m+1)/2$ [13]. Since $3\,A051403(x) + (x+3)\,A051403(x-1) = X$, the proof above also confirms this formula for $n \geq 5$; this identification is not part of the formal proof.

3.3. **Three arithmetic lemmas.** These lemmas are also used in [8]; we include their short proofs.

Lemma 8. *For integers $q \geq 1$ and $k \geq 0$, $q^k \mid (qk)!$.*

Proof. Induction on k: $(q(k+1))! = q(k+1) \cdot (qk + q - 1)!$, where $q \mid q(k+1)$ and $(qk)! \mid (qk + q - 1)!$. □

Lemma 9 (lifting)**.** *Let $N \geq 1$ and $M, q \in \mathbb{N}$, put $g = \gcd(N, M)$ and $d = N/g$, and suppose $q \mid d$. Then for every $j \geq 0$, $q^j \mid M$ implies $q^{j+1} \mid N$.*

Proof. Note $N = gd$. For $j = 0$, $q \mid d \mid N$. If the claim holds for j and $q^{j+1} \mid M$, then $q^j \mid M$, so $q^{j+1} \mid N$; hence $q^{j+1} \mid \gcd(N, M) = g$, and $q^{j+2} = q^{j+1} \cdot q \mid gd = N$. □

Lemma 10. *If $q \geq 3$, $j \geq 3$ and $qj \geq 34$, then $(qj+4)^2 < q^j$.*

Proof. If $j = 3$ then $q \geq 12$, and $q^3 \geq 12q^2 \geq 9q^2 + 36q > (3q+4)^2$. If $j = 4$ then $q \geq 9$, and $q^4 \geq 81q^2 > 32q^2 \geq 16(q+1)^2 = (4q+4)^2$. If $j = 5$ then $q \geq 7$, and $q^5 \geq 343q^2 > 36q^2 \geq (5q+4)^2$. For $j \geq 6$ we induct on j, for any $q \geq 3$: $q^6 \geq 81q^2 > 64q^2 \geq (6q+4)^2$, and if $(qj+4)^2 < q^j$ with $j \geq 6$, then $q \leq (qj+4)/6$, so $(q(j+1)+4)^2 \leq \left(\frac{7}{6}\right)^2 (qj+4)^2 < 3q^j \leq q^{j+1}$. □

3.4. **The main case.**

Proposition 11. *Let $x \geq 33$, $g = \gcd(N, M)$ and $d = N/g$. Then $d = 1$ or d is prime.*

Proof. We have $N = gd > 0$, $d \mid N$, $x! \mid M$, and

$$(x+3)^2 < N < (x+4)^2,$$

since $N - (x+3)^2 = x$ and $(x+4)^2 - N = x + 7$. Suppose $d \neq 1$, and let p be the least prime factor of d. Then $p \mid N$, so $p \geq 3$ as N is odd.

Case (A): $2p \le x$. Let $k = \lfloor x/p \rfloor \ge 2$. Since $pk \le x$, Lemma 8 gives $p^k \mid (pk)! \mid x! \mid M$, so $p^{k+1} \mid N$ by Lemma 9, and $p^{k+1} \le N$. Moreover $x < p(k+1)$, so $p(k+1) \ge 34$ and $x + 4 \le p(k+1) + 4$. Lemma 10 with $q = p$ and $j = k + 1 \ge 3$ gives $p^{k+1} \le N < (x+4)^2 \le \big(p(k+1)+4\big)^2 < p^{k+1}$, a contradiction.

Case (B): $p \le x < 2p$. Then $p \mid x! \mid M$, so $p^2 \mid N$ by Lemma 9; write $N = mp^2$ with $m \ge 1$. If $m \ge 5$, then, as $2p \ge x + 1$, $4N = m(2p)^2 \ge 5(x+1)^2 = 4N + x(x-18) - 31 > 4N$, which is absurd. So $1 \le m \le 4$, and $m \ne 2, 4$ as N is odd. If $m = 1$, then $(x+3)^2 < p^2 = N < (x+4)^2$, so $x + 3 < p < x + 4$, which is impossible. Hence $N = 3p^2$. As $N > 27$, $p \ne 3$, and $3 \mid M$ by the definition of M; since also $p \mid M$, the distinct primes 3 and p give $3p \mid M$. Also $3p \mid N$, hence $3p \mid g$; write $g = 3ph$. Then $3p \cdot p = N = gd = 3p \cdot hd$ gives $p = hd$, so $d \mid p$ and $d = p$ is prime.

Case (C): $p > x$. We have

$$N = (x+1)(x+6) + 3, \quad N + 1 = (x+2)(x+5), \quad N + 3 = (x+3)(x+4).$$

So if $x + i$ divided N for some $i \in \{1, 2, 3\}$, it would divide 3, 1 or 3 respectively, which is impossible as $x + 1 \ge 34$. Hence every divisor of N exceeding x is at least $x + 4$; in particular $p \ge x + 4$. Suppose $d \ne p$, and let $r = d/p$. Then $r \ne 1$, and $r \ne 0$ as $d \ne 0$. The least prime factor q of r divides d, so $q \ge p$ by minimality of p. Hence $r \ge q \ge p \ge x + 4$ and $d = pr \ge (x+4)^2 > N \ge d$, a contradiction. So $d = p$ is prime. □

Remark 12. For $x = 2$ ($n = 5$) we have $N = 27 = 3^3$, $M = 24 = 2^3 \cdot 3$, $g = 3$ and $d = 9$: the least prime factor $p = 3 = x+1$ of d exceeds x, and the step "$x+1 \mid 3$ is impossible" of Case (C) fails. In Case (B) the factor 3 of M, which comes from the endpoint -3, is what makes the case $N = 3p^2$ harmless. The threshold $x \ge 33$ is chosen for convenience in Cases (A) and (B).

3.5. **Proof of Theorem 2 and Corollary 3.** Let $n \ge 3$. If $n \ge 36$, then $x = n - 3 \ge 33$, and $a(n) = N/\gcd(N, M)$ is 1 or prime by Propositions 7 and 11. If $3 \le n \le 35$, then $a(n)$ is one of the first 33 entries of the list in `a_data`, which equals $(a(3), \dots, a(59))$ by kernel computation; each of its 57 entries is 9, 1 or a prime (the largest is 2753). This proves Theorem 2. The only entry of the list that is neither 1 nor prime is $a(5) = 9$, which gives Corollary 3; it is not a separate Lean theorem, but follows at once from `a_eq`, `main` and `a_data`. □

4. Machine verification

Files and declarations. The proof is the file `lean/Math0/Frontier/OEIS/A362086.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A362086.result`, of type `Math0.Statements.A362086.statement`. Besides the private declarations named in Section 3, the file contains `lf`, `lf_succ`, `two_dvd_lf` (left factorial), `fold_concat` (peeling off the innermost level of the fold), `pow_dvd_fact`, `pow_step`, `bound6`, `bound` (Lemmas 8–10), `not_dvd_aux`, `main` (Proposition 11) and `small` (the cases $3 \le n < 60$, from `a_data`; only $n < 36$ is used). Here `main` states that $N/\gcd(N, M)$ is 1 or prime for $x \ge 33$, and the final proof reads, verbatim:

```
theorem result : statement := by
  intro n hn h9
  by_cases hsmall : n < 36
  · exact small n hn (by omega) h9
  · obtain ⟨x, rfl⟩ : ∃ x, n = x + 3 := ⟨n - 3, by omega⟩
    rw [a_eq x (by omega)]
    exact main x (by omega)
```

Kernel computation. In the statement file, `a_data` and the examples are proved by `decide +kernel`: the Lean kernel evaluates the continued fractions for $n = 3, \dots, 59$ in exact rational arithmetic and compares the reduced denominators with the 57 OEIS terms (about 6 seconds of kernel time). In the proof file, `small` rewrites with `a_data` and discharges the 57 claims "$v \neq 9 \Rightarrow v = 1 \vee v$ prime" with `norm_num`, whose primality extension produces proof terms that the kernel checks; the base case $2 \mid !\,2$ of `two_dvd_lf` is closed by `decide`. Neither file uses `native_decide` (which would add the axiom `Lean.ofReduceBool`) or `sorry`, and the proof file sets no options.

Audit. The file ends with the command `#math0_settles` applied to `result` and `statement`. This command, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a definition of type `Prop` without universe parameters; (2) `statement` and `result` depend on no axioms other than `propext`, `Classical.choice` and `Quot.sound` (so no `sorryAx`, `Lean.ofReduceBool` or custom axiom); and (3) the type of `result` is syntactically the statement or its negation, up to metadata annotations, with no unfolding. Running `lake env lean Math0/Frontier/OEIS/A362086.lean` in `lean/` prints

```
math0 audit: 'Math0.Statements.A362086.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A362086.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A362086.result' proves 'Math0.Statements.A362086.statement'
```

Toolchain and reproduction. We use Lean 4 [10] `v4.33.1` and Mathlib [9] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [5] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib cache, rebuilds all math0 results with `lake build` (re-running every audit command) and checks that the pinned dependencies are unmodified.

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Produced by Claude (an AI model developed by Anthropic) in the math0 project